

Exchange Rate Regimes in Production Networks

A Small Open Economy Extension of Rubbo (2024)

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Building on: Rubbo (2024) *Networks, Phillips Curves, and Monetary Policy*

Outline

Intro

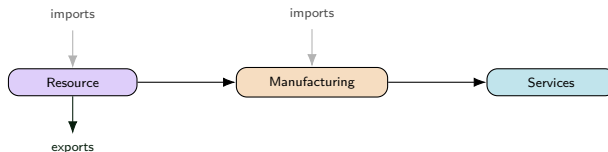
Model

Results

Conclusion

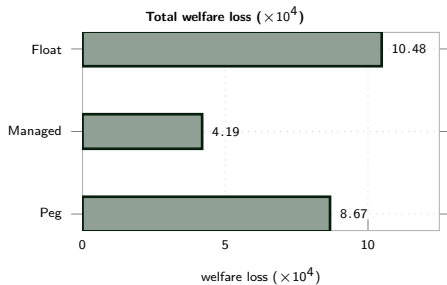
Motivation

- Network Phillips curves (Rubbo 2024, La'O–Tahbaz-Salehi 2022): **closed economy**, unique DC index
- Open-economy NK (Galí–Monacelli 2005): FX channel, **no network**
- **This paper: both together** — multi-sector network + FX



Questions: (1) which FX regime is welfare-optimal? (2) does the DC index survive a second (FX) cost-push channel?

Headline Result



- **Managed float dominates** both corners
- Float targets DC index directly — still **not** best
- Driver: ER **persistence**, not volatility (Welfare section)

Literature

- **La'O & Tahbaz-Salehi (2022)** — optimal MP, production networks, closed economy
- **Rubbo (2024)** — base paper: Calvo network $\Rightarrow$ unique DC index
- **Auer, Levchenko & Sauré (2019)** — inflation spillovers via global IO
- **Pasten, Schoenle & Weber (2020)** — sectoral rigidity heterogeneity, transmission
- **Fanelli & Straub (2021)** — FX intervention theory
- **Gopinath & Itskhoki (2022); Amiti et al. (2019)** — dominant currency, pass-through

Gap: network + open economy + FX regimes — together, no one has this.

This Paper

- Extend Rubbo → SOE: imported inputs Ω^F , UIP, debt-elastic premium
- Ask: optimal FX regime? DC index survive?
- Calibrate: 3-sector oil exporter
- Solve **nonlinear** model directly (not hand-linearized)

Model: Households

$$\begin{aligned} \max \mathbb{E}_0 \sum_t \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right] \quad \text{s.t.} \quad & P_t^C C_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i^*) \left[1 - \psi(\hat{b}_{t-1}^* - \bar{b}^*) \right] S_t B_{t-1}^* \\ & C_t^{-\gamma} = \beta(1+i_t) \mathbb{E}_t \left[C_{t+1}^{-\gamma} / \Pi_{t+1}^C \right], \quad W_t / P_t^C = C_t^\gamma L_t^\varphi \\ \text{UIP (derived):} \quad & (1+i_t) = (1+i^*) \left[1 - \psi(\hat{b}_t^* - \bar{b}^*) \right] \mathbb{E}_t[S_{t+1}/S_t] \end{aligned}$$

C, L : consumption, labor
 ψ : debt-elastic premium

β, γ, φ : discount, RRA, inv. Frisch
 B^* : foreign bond

W, i : wage, dom. rate
 P^C : CPI

S, i^* : ER, foreign rate
 Π_t : profits (rebated)

Model: Firms — Technology and Cost

$$Y_{it} = A_{it} L_{it}^{\alpha_i} \prod_j X_{ijt}^{\Omega_{ij}^H} M_{it}^{\Omega_i^F} \quad \Rightarrow \quad MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

$$W_t L_{it} = \alpha_i MC_{it} Y_{it}, \quad P_{jt} X_{ijt} = \Omega_{ij}^H MC_{it} Y_{it}, \quad S_t P_t^{F*} M_{it} = \Omega_i^F MC_{it} Y_{it}$$

Y, A : output, TFP L, X, M : labor, dom./imp. inputs Ω_{ij}^H : dom. IO share Ω_i^F : import cost share
 α_i : labor share MC_{it} : nominal marg. cost P_t^{F*} : foreign import price

Key: MC_{it} moves with both A_{it} (TFP) and S_t (via M_{it}) — the two cost-push channels.

Model: Firms — Calvo Pricing

Reset price (probability δ_i resets, $1-\delta_i$ keeps last price):

$$X1_{it} = \widetilde{MC}_{it} Y_{it} + (1-\delta_i)\beta \left(\frac{C_{t+1}}{C_t}\right)^{-\gamma} \Pi_{i,t+1}^\varepsilon X1_{i,t+1}, \quad X2_{it} = Y_{it} + (1-\delta_i)\beta \left(\frac{C_{t+1}}{C_t}\right)^{-\gamma} \Pi_{i,t+1}^{\varepsilon-1} X2_{i,t+1}$$

$$P_{it}^\# / P_{it} = \frac{\varepsilon}{\varepsilon-1} X1_{it} / X2_{it}, \quad 1 = (1-\delta_i)\Pi_{it}^{\varepsilon-1} + \delta_i \left(P_{it}^\# / P_{it}\right)^{1-\varepsilon}$$

δ_i : Calvo reset prob. (primitive) $\hat{\delta}_i$: derived slope $P_{it}^\#$: reset price ε : elasticity (markup $\varepsilon/(\varepsilon-1)$)

$\hat{\delta}_i = \frac{\delta_i(1-\beta(1-\delta_i))}{1-\beta\delta_i(1-\delta_i)}$ — appears in DC weights (Theory section), *derived* not primitive.

Model: Open Economy

$$\underbrace{P_t^{FH} = S_t P_t^{F*}}_{\text{LOP}}, \quad \underbrace{(1+i_t) = (1+i^*) \left[1 - \psi(\hat{b}_t^* - \bar{b}^*) \right] \mathbb{E}_t[S_{t+1}/S_t]}_{\text{UIP}}$$

$$\hat{b}_t^* = \frac{1+i_{t-1}}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H EX_t - P_t^{FH} IM_t}{P_t^C}, \quad EX_t = \kappa_{EX} D_t^* \left(\frac{P_t^H}{P_t^{FH}} \right)^{-\theta^*}$$

$$Y_{it} = \beta_i^H \frac{P_t^H}{P_{it}} \left(C_t^H + EX_t \right) + \sum_j \Omega_{ji}^H \frac{MC_{jt}}{P_{it}} \frac{Y_{jt}}{P_{it}}$$

$\hat{b}^*$: NFA (scaled)

β_i^H : cons. share, sector i

EX, IM : exports, imports

Float: i_t Taylor rule, S_t free

θ^* : export elasticity

D^* : foreign demand shifter

Peg: $S_t = \bar{S}$, i_t residual

Model: Policy Regimes and Shocks

Regime	Rule	Exchange rate
Float	$i_t = \bar{i} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$	fully endogenous (UIP)
Peg	$S_t = \bar{S}$	fixed; i_t residual
Managed	$i_t = \bar{i} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \log(S_t/\bar{S})$	partially stabilized

Shocks (AR(1)): a_{it} (TFP, $\rho=0.90$) p_t^{F*} (import price, $\rho=0.85$) D_t^* (foreign demand, $\rho=0.80$)

Theory: Regime Comparison — Predictions

	Float	Peg	Managed
Exchange rate	endogenous	fixed	partial
FX cost-push absorbed by	S_t	π_i, y	split
Monetary independence	full	none	full

Naive: float best (targets π_t^{DC} , well-defined by construction — see appendix). **Overtaken** — see Welfare section.

Calibration: 3-Sector Oil-Exporting Economy

Parameter	Value	Parameter	Value
β	0.99	$\Omega_{21}^H, \Omega_{32}^H$	0.20, 0.25
γ, φ	1.00, 2.00	$\Omega_1^F, \Omega_2^F, \Omega_3^F$	0.30, 0.10, 0.05
η	1.50	$\beta_1^H, \beta_2^H, \beta_3^H$	0.05, 0.15, 0.80
ε	8.00	$\delta_1, \delta_2, \delta_3$	0.75, 0.50, 0.25
θ^*	2.00	ψ	0.020
$\beta^{F,tot}$	0.10	ϕ_π, ϕ_y, ϕ_s	1.50, 0.50, 0.30

Resource: upstream, flexible, import/export-heavy → Manufacturing: mid-chain → Services: sticky, consumption-dominant.

Solved: nonlinear, order 1; steady state $\Rightarrow$ scalar root-find, $W^* = 1.3189$.

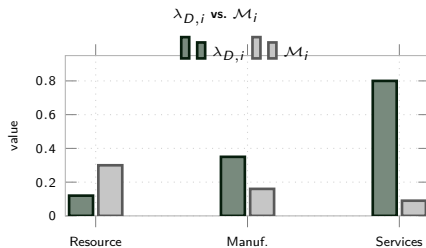
Calibration: Network Properties

$$\lambda_{D,i} = \left[(\beta^H)^\top (I - \Omega^H)^{-1} \right]_i$$

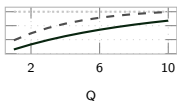
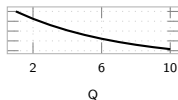
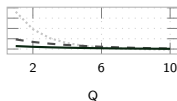
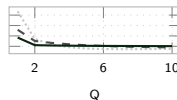
$$\mathcal{M}_i = \left[(I - \Omega^H)^{-1} \Omega^F \mathbf{1} \right]_i$$

$$w_i^{DC} \propto \lambda_{D,i} \frac{1 - \hat{\delta}_i}{\hat{\delta}_i}$$

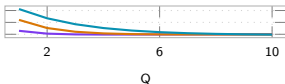
$\lambda_{D,i}$: Domar weight $\mathcal{M}_i$: import centrality w_i^{DC} : DC weight



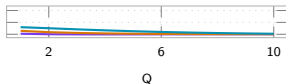
IRFs: Import-Price Shock

Exch. rate S_t (% , dep.>0)Shock: import price P_t^{F*} (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

Sectoral output gap, Peg (%)



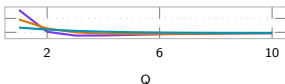
Sectoral output gap, Managed (%)



Sectoral output gap, Float (%)



Sectoral inflation, Peg (%)



Sectoral inflation, Managed (%)

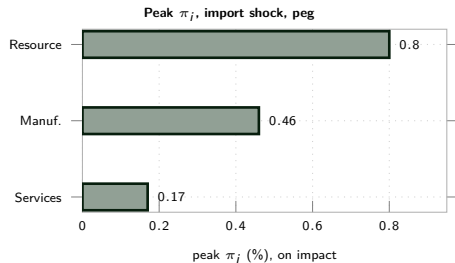
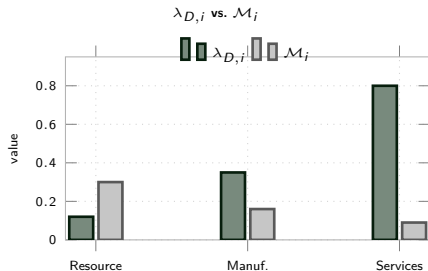


Sectoral inflation, Float (%)



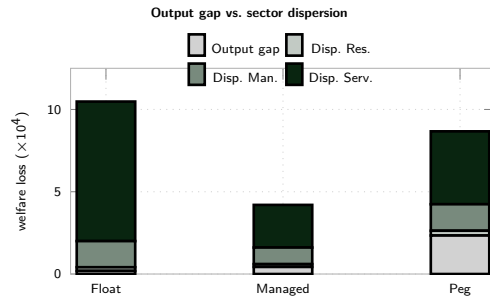
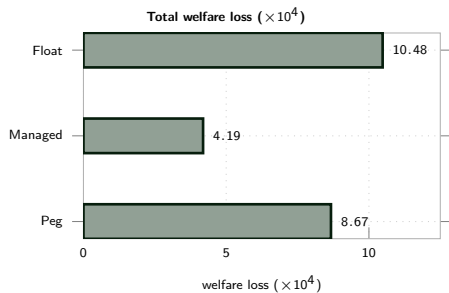
Peg ■ ■ ■ Managed - - Float — Shock itself —. By sector (Float): — Res. — Man. — Serv.

Network: Centrality vs. Exposure



Resource: high $\mathcal{M}_i$ (0.30), low $\lambda_{D,i}$ (0.12) $\rightarrow$ volatile locally, small aggregate. **Services:** low $\mathcal{M}_i$ (0.09), high $\lambda_{D,i}$ (0.80) $\rightarrow$ dominates aggregate.

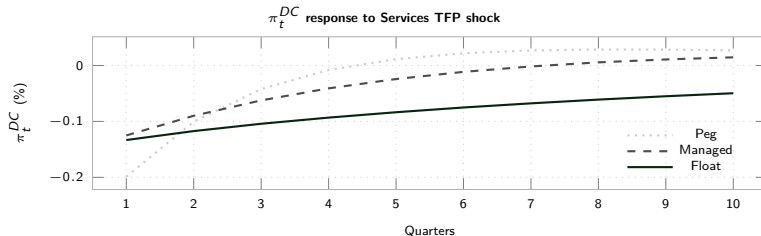
Welfare: Headline



$\mathcal{W} = \frac{1}{2} \left[(\gamma + \varphi) \text{Var}(\tilde{y}_t) + \sum_i \lambda_{D,i} \varepsilon \frac{1 - \delta_i}{\delta_i} \text{Var}(\pi_{it}) \right]$ — Float: lowest output-gap cost, largest Services dispersion; Managed: not dominated on any margin.

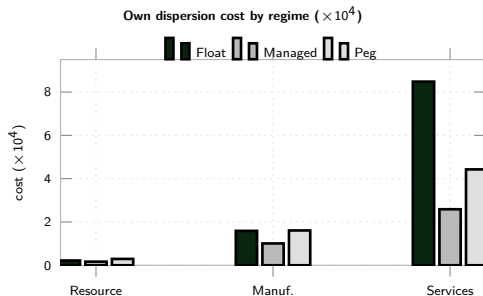
Welfare: Persistence, Not Volatility

Under float, S_t is pinned by UIP + near-unit-root NFA ($AR \approx 0.994$) $\Rightarrow$ feeds every sector's MC via Ω_i^F , even for pure domestic shocks.



Peg: fast, one-time overshoot/reversal. Float: barely decays — **no FX shock here**, pure domestic TFP.

Welfare: Sector Preferences

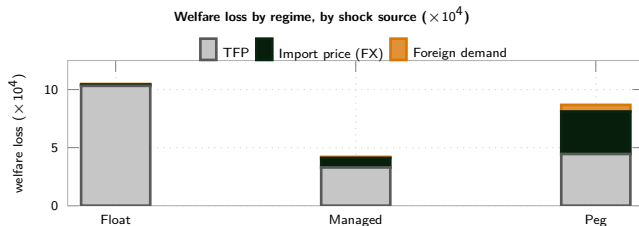


$$\Gamma_i = \frac{[\Delta(I - \Omega\Delta)^{-1}\omega_F]_i}{\kappa}, \quad \kappa_i = \lambda_{D,i} \varepsilon \frac{1 - \hat{\delta}_i}{\hat{\delta}_i}$$

	Res.	Man.	Serv.
Γ_i	0.234	0.054	0.006
κ_i	0.43	5.54	74.6

- All: **Managed** first choice
- Resource (high Γ): Float > Peg
- Services (high κ): Peg > Float

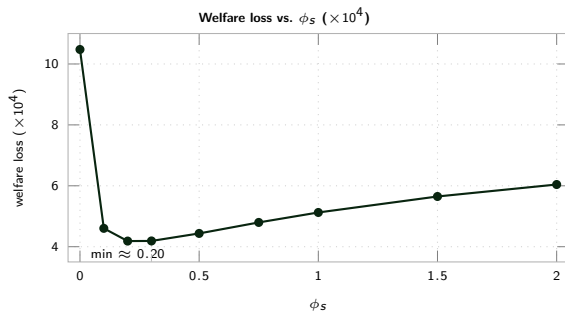
Shock Decomposition of Welfare Cost



- **TFP dominates**, not FX, for Float & Managed (10.32/10.48; 3.29/4.19)
- **FX channel matters only for Peg** (3.66, $\approx 42\%$ of total)
- Float: FX cost ≈ 0 (insulated) but TFP persistence high

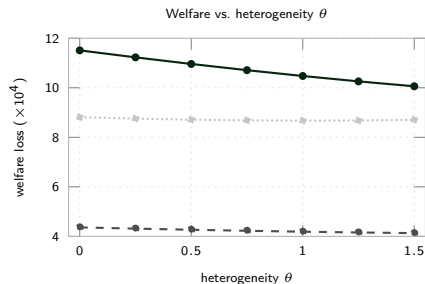
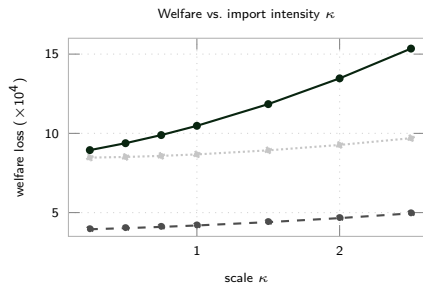
Chart above: original 5-shock model. Export-price/ToT (ε_{pX}) and foreign risk-premium/UIP (ε_{rp}) shocks are now implemented and simulated (not yet folded into this chart): ε_{rp} alone explains $\approx 78\%$ of Peg's π_t^{DC} variance (zero monetary autonomy) and would raise Peg's total welfare loss well above Float's — reverses part of the headline ranking. Full 7-shock re-numbering across all slides: Next Steps.

Generalization: Optimal FX Stabilization ϕ_s



- $\phi_s=0 \rightarrow 0.10$: loss $10.48 \rightarrow 4.60$
- Min flat, wide: $\phi_s \in [0.2, 0.3]$
- Calibrated $\phi_s=0.30$ near-optimal
- Beyond ≈ 0.3 : cost rises again

Generalization: Robustness



Managed < **Peg** < **Float** robust across $\kappa \in [0.25, 2.5]$ and $\theta \in [0, 1.5]$. Heterogeneity gap **shrinks** with θ — opposite naive $\Gamma \propto \mathcal{B}$ prediction (α_i absorbs Ω^F changes).

Conclusion

- **Managed float optimal:** loss 4.19 vs. Float 10.48, Peg 8.67 ($\times 10^4$)
- **Driver: ER persistence** (UIP + near-unit-root NFA), not on-impact volatility
- **Sectors split:** Resource $\rightarrow$ Float $>$ Peg; Services $\rightarrow$ Peg $>$ Float
- **Optimal** $\phi_s \approx 0.2\text{--}0.3$: small, strictly positive
- **Robust** to import intensity κ and heterogeneity θ
- First-order welfare only (order-2 fails: price-level/ER unit root)
- **Next:** price-level anchor (2nd-order welfare); **propagate new 7-shock welfare numbers** (ToT ε_{pX} , risk-premium ε_{rp} now simulated — Peg's loss rises sharply, Managed still wins) across all slides; δ_i, Ω^H sensitivity

Appendix: DC Index — Definition and Insulation

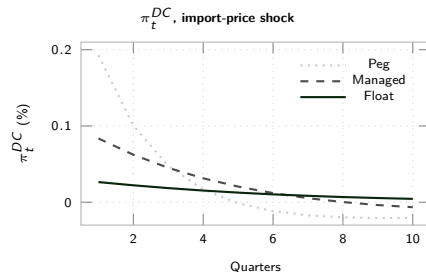
Closed economy (Rubbo): 1 channel (TFP), $\mathcal{V}$
rank-deficient $\Rightarrow$ unique DC index:

$$\phi^\top \mathcal{V} = \mathbf{0}^\top \Rightarrow \pi_t^{DC} = \sum_i w_i^{DC} \pi_{it}, \quad w_i^{DC} \propto \lambda_{D,i} \frac{1-\delta_i}{\hat{\delta}_i}$$

Open economy: 2 channels (TFP + FX):

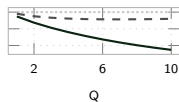
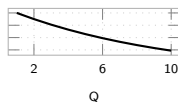
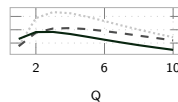
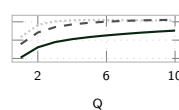
$$\underbrace{\phi^\top \mathcal{V} = \mathbf{0}^\top}_{\text{TFP}} \text{ and } \underbrace{\phi^\top \Gamma = 0}_{\text{FX}} \Rightarrow \text{generically infeasible}$$

Fix: import nodes flexible ($\hat{\delta}_M=1 \Rightarrow w_M=0$) $\Rightarrow$ DC well-defined again by construction. This is what the Taylor rules target throughout (π_t^{DC}); FX insulation itself is a quantitative question, shown at right.

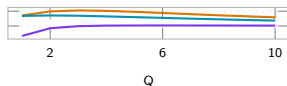


Float: smallest on impact (0.026%) & most persistent;
Peg/Managed overshoot and reverse sign — on-impact insulation $\neq$ full story.

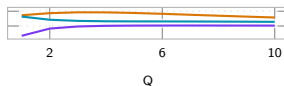
IRFs: Resource TFP Shock

Exch. rate S_t (% , dep.>0)Shock: TFP A_{1t} (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

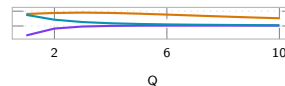
Sectoral output gap, Peg (%)



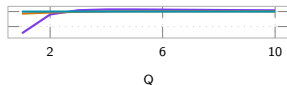
Sectoral output gap, Managed (%)



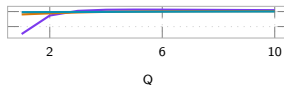
Sectoral output gap, Float (%)



Sectoral inflation, Peg (%)



Sectoral inflation, Managed (%)

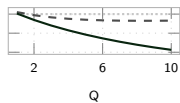
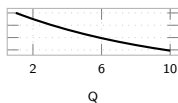
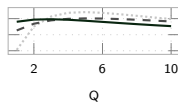
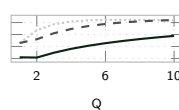


Sectoral inflation, Float (%)

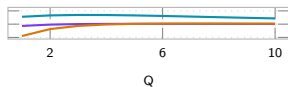


Peg ■ ■ ■ Managed - - Float — Shock itself —. By sector (Float): — Res. — Man. — Serv.

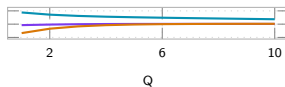
IRFs: Manufacturing TFP Shock

Exch. rate S_t (% , dep.>0)Shock: TFP A_{2t} (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

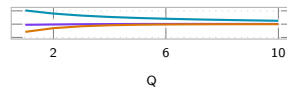
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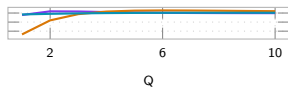
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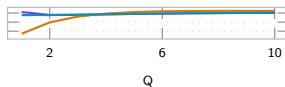
Sectoral output gap, Float (%)



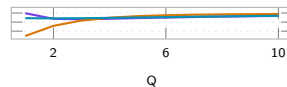
Sectoral inflation, Peg (%)



Sectoral inflation, Managed (%)

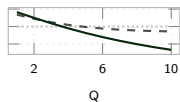
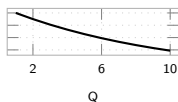
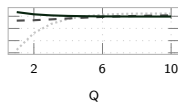
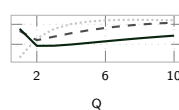


Sectoral inflation, Float (%)

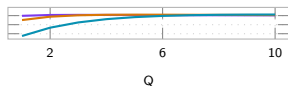


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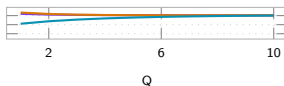
IRFs: Services TFP Shock

Exch. rate S_t (% , dep.>0)Shock: TFP A_{3t} (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

Sectoral output gap, Peg (%)



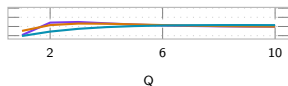
Sectoral output gap, Managed (%)



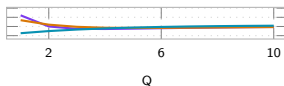
Sectoral output gap, Float (%)



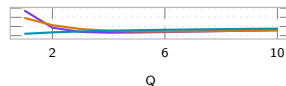
Sectoral inflation, Peg (%)



Sectoral inflation, Managed (%)

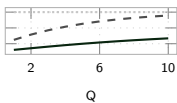
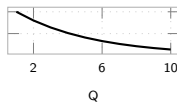
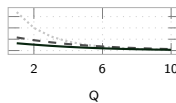
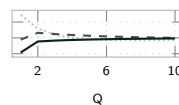


Sectoral inflation, Float (%)

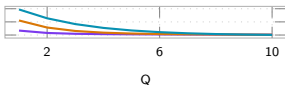


Peg ■ ■ ■ Managed - - Float — Shock itself — By sector (Float): — Res. — Man. — Serv.

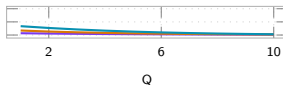
IRFs: Foreign-Demand Shock

Exch. rate S_t (% , dep.>0)Shock: foreign demand D_t^* (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

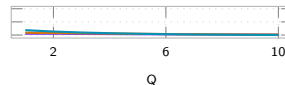
Sectoral output gap, Peg (%)



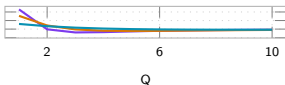
Sectoral output gap, Managed (%)



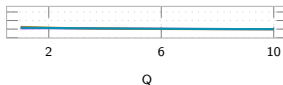
Sectoral output gap, Float (%)



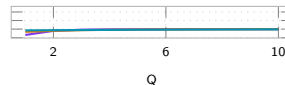
Sectoral inflation, Peg (%)



Sectoral inflation, Managed (%)

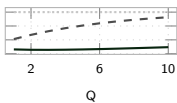
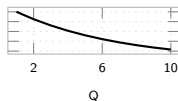
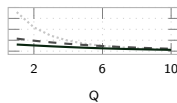
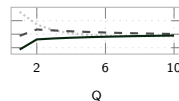


Sectoral inflation, Float (%)

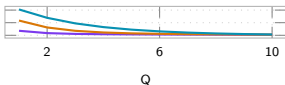


Peg ■ ■ ■ Managed - - Float — Shock itself — By sector (Float): — Res. — Man. — Serv.

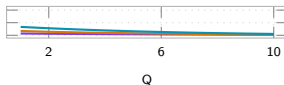
IRFs: Export-Price / ToT Shock

Exch. rate S_t (% , dep.>0)Shock: export price P_t^X (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

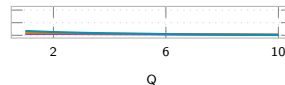
Sectoral output gap, Peg (%)



Sectoral output gap, Managed (%)



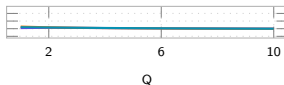
Sectoral output gap, Float (%)



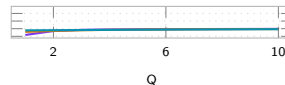
Sectoral inflation, Peg (%)



Sectoral inflation, Managed (%)

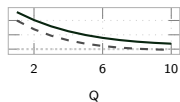
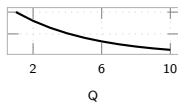
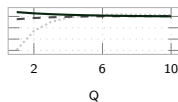
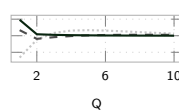


Sectoral inflation, Float (%)

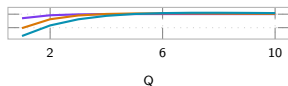


Peg ■ ■ ■ Managed - - Float — Shock itself — By sector (Float): — Res. — Man. — Serv.

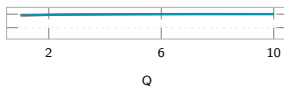
IRFs: Foreign Risk-Premium (UIP) Shock

Exch. rate S_t (% , dep.>0)Shock: risk premium RP_t (%)Output gap $\tilde{y}_t$ (%)CPI infl. π_t^C (%)

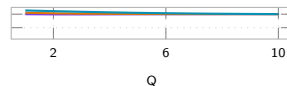
Sectoral output gap, Peg (%)



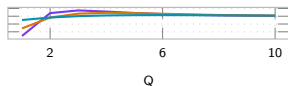
Sectoral output gap, Managed (%)



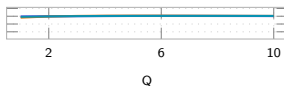
Sectoral output gap, Float (%)



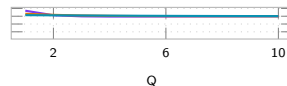
Sectoral inflation, Peg (%)



Sectoral inflation, Managed (%)



Sectoral inflation, Float (%)



Peg ■ ■ ■ Managed - - Float — Shock itself — By sector (Float): — Res. — Man. — Serv.